

MESSSTETTEN, GermanyWMO: **108270**Lat: **48.1694N**Lon: **8.9433E**Elev: **2954**StdP: **13.19**Time Zone: **1.00 (EUC)**Period: **94-12**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	13.9	6.2	8.4	16.9	10.2	10.3	18.3	23.3	39.7	20.8	38.1	5.3	70	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	80.3	63.1	77.0	61.9	73.9	61.0	65.0	75.9	63.5	74.0	62.0	71.8	5.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	89.7	68.9	59.4	84.5	67.3	58.2	80.9	66.0	31.6	75.9	30.4	74.2	29.3	72.0	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.4	14.5	12.3		5.4	85.1	4.4	2.9	2.2	87.2	-0.4	88.9	-2.9	90.6	-6.1	92.7	
			DB	5.2	67.9	4.5	1.9	2.0	69.3	-0.7	70.4	-3.2	71.4	-6.5	72.8	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	30.0	31.6	37.4	44.2	52.3	57.9	60.7	60.8	53.5	46.8	37.3	30.9
	DBStd	13.56	8.08	8.92	8.31	8.02	7.26	7.24	6.44	6.55	6.71	7.57	8.02	7.96
	HDD50	2988	621	517	395	208	60	14	2	3	40	152	384	592
	HDD65	7259	1086	936	857	624	398	230	167	163	347	564	830	1057
	CDD50	1292	0	0	4	34	130	251	333	338	145	52	5	0
	CDD65	88	0	0	0	0	3	17	32	33	3	0	0	0
	CDH74	664	0	0	0	1	23	130	226	268	16	0	0	0
	CDH80	104	0	0	0	0	0	12	34	58	0	0	0	0
Wind	WSAvg	5.8	6.1	6.5	6.6	5.9	5.7	5.3	5.3	4.9	5.3	5.6	5.7	6.1
Precipitation	PrecAvg	35.4	2.2	2.2	2.3	2.5	3.9	3.9	4.1	3.3	2.7	2.8	2.7	2.7
	PrecMax	44.6	5.5	5.4	5.9	4.8	5.9	6.7	8.3	6.0	5.0	6.0	5.1	6.1
	PrecMin	26.4	0.2	0.7	0.6	0.3	1.7	1.8	2.0	0.7	1.0	0.4	0.1	0.0
	PrecStd	4.3	1.3	1.1	1.3	1.2	1.4	1.2	1.3	1.2	1.1	1.3	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	55.2	62.2	69.5	77.3	82.1	83.8	86.1	76.2	69.7	59.7	50.4
		MCWB	40.7	42.8	48.0	52.6	59.1	63.3	64.1	63.5	60.4	58.0	48.9	43.6
	2%	DB	46.4	50.3	57.2	65.7	72.8	77.8	79.7	80.4	71.5	64.8	54.8	46.1
		MCWB	41.3	41.3	45.5	50.5	57.9	62.3	63.3	63.3	59.3	55.8	47.3	41.3
	5%	DB	43.7	46.4	53.0	61.5	68.8	74.5	76.1	76.6	67.9	60.8	51.1	43.8
		MCWB	39.7	39.6	43.8	48.9	55.6	61.3	62.5	61.9	57.6	53.4	45.8	40.4
	10%	DB	40.8	43.0	49.1	57.5	65.2	71.0	72.4	72.7	64.2	57.3	48.1	41.3
		MCWB	37.4	38.2	42.7	47.3	54.0	59.9	61.0	60.9	56.0	51.9	44.2	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	45.0	49.6	54.0	62.0	66.5	66.7	66.7	63.5	59.3	50.9	45.3
		MCDB	47.0	51.7	57.8	66.2	72.1	76.1	77.7	78.7	71.5	66.8	56.2	47.8
	2%	WB	42.4	42.5	47.2	51.7	59.3	63.9	64.9	64.7	61.2	56.7	48.4	42.9
		MCDB	45.2	47.7	53.6	63.2	69.6	74.2	75.8	76.4	68.4	62.0	52.9	45.0
	5%	WB	40.1	40.6	45.2	49.6	56.9	62.1	63.4	63.0	59.0	54.7	46.6	40.9
		MCDB	43.3	45.5	50.8	59.6	66.8	72.0	73.9	73.9	65.4	59.2	50.4	43.1
	10%	WB	37.5	38.5	43.2	47.8	55.0	60.4	61.8	61.4	57.0	52.6	44.6	39.1
		MCDB	40.2	42.3	48.4	56.5	63.4	69.6	71.2	71.4	63.0	56.6	47.4	41.0
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.4	13.0	15.1	16.1	16.4	16.4	16.3	14.6	13.5	10.4	9.2
		MCDBR	13.8	16.5	18.5	20.5	21.7	21.3	21.5	22.2	19.8	18.7	15.7	12.6
		MCWBR	9.7	10.3	10.6	10.4	11.1	10.7	9.9	10.5	10.2	10.6	9.8	9.4
	5% WB	MCDBR	12.3	14.3	15.9	18.8	19.4	19.4	19.5	20.1	17.6	16.7	13.8	10.9
		MCWBR	9.5	9.6	9.8	10.0	10.7	10.1	9.6	10.2	9.6	10.2	9.2	9.1
Clear-Sky Solar Irradiance	taub	0.279	0.301	0.332	0.363	0.373	0.377	0.377	0.373	0.350	0.327	0.299	0.276	
	taud	2.478	2.423	2.390	2.332	2.332	2.347	2.335	2.367	2.416	2.474	2.505	2.478	
	Ebn at Noon	257	274	281	282	282	280	279	275	271	260	246	243	
	Edh at Noon	20	26	31	37	38	38	38	35	30	24	19	17	
All-Sky Solar Radiation	RadAvg	326	576	955	1352	1584	1815	1717	1501	1116	688	371	263	
	RadStd	40	89	135	194	166	153	153	149	118	74	47	44	

Historical Trends

	DBAvg	Heating	
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